

A $(3.49\dots)^k$ bound for diagonal Ramsey numbers

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Abstract

The diagonal Ramsey number $R(k, k)$ is the least order of a complete graph such that every red/blue colouring of its edges contains a monochromatic clique of order k . We prove $R(k, k) \leq b_*^k$ for all sufficiently large k , where $b_* = 3.49862379380751\dots$ is an explicit constant. In particular, $R(k, k) \leq 3.4986238^k < 3.5^k$ for all sufficiently large k , improving the bound 3.69507^k of Lu and Wang. The classical bound $R(k, k) < 4^k$ was proved by Erdős and Szekeres in 1935. Its exponential base was first reduced by Campos, Griffiths, Morris and Sahasrabudhe in 2023. Our proof strengthens their algorithm by tracking a potential that gives greater weight to vertices with high red density. An explicit integral certifies each use of the resulting local step. We then use the improved off-diagonal bounds as inputs to further applications of the method. Induction on the number of vertices justifies these dependencies without assuming the bounds to be proved. The finite numerical inequalities are certified with exact arithmetic, and the complete argument, including the off-diagonal input, has been formalised in Lean.

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1 Introduction

For positive integers k and l , the Ramsey number $R(k, l)$ is the least N such that every colouring of the edges of the complete graph K_N in red and blue contains a red K_k or a blue K_l . Erdős and Szekeres [2] proved $R(k, k) \leq \binom{2k-2}{k-1} < 4^k$ in 1935. For almost ninety years the base 4 was not improved. Campos, Griffiths, Morris and Sahasrabudhe [1] then showed $R(k, k) \leq (4 - \varepsilon)^k$ for a small fixed $\varepsilon > 0$, by a new algorithm that builds large monochromatic “books”. Gupta, Ndiaye, Norin and Wei [3] optimised their method to reach about 3.8^k . Lu and Wang [4] recently obtained

$$R(k, k) \leq 3.69507^k \quad \text{for all large } k,$$

with a computer-assisted proof that is also checked in Lean.

Our main result is the following.

Theorem 1.1. *Put*

$$z^* = \frac{626184844701}{500000000000} = 1.252369689402, \quad b_* = e^{z^*}.$$

There is an integer $K \geq 1$ such that, for every integer $k \geq K$,

$$R(k, k) \leq b_*^k \leq \left(\frac{34986238}{10000000} \right)^k < (7/2)^k.$$

Here $b_ = 3.49862379380751 \dots$ and $34986238/10000000 = 3.4986238$. There is also an absolute constant $C \geq 1$ such that $R(k, k) \leq C e^{z^*k}$ for every $k \geq 1$.*

The notation $3.49 \dots$ in the title denotes b_* , not the smaller decimal 3.49 . The proof gives a profile with exponent strictly below z^* . This strict gap lets us absorb the constant factor and obtain the eventual bound with base b_* itself. The decimal upper bound 3.4986238 is certified separately. The constant C and the threshold K are not effective, because Theorem 3.4 uses a compactness argument.

Formal verification and data. We formalise the proof in Lean 4 [8] using Mathlib [7], Lu and Wang’s development [5], and its dependency `RamseyLean` [6]. Section 7 explains the numerical checks, the modular Lean build, and the final statement and axiom checks. The paper, certificate and reproduction instructions will be available at `[repository URL to be inserted]`.

What is new. The proof follows the book algorithm of Campos, Griffiths, Morris and Sahasrabudhe [1], in the finite form developed by Lu and Wang [4]. A monochromatic *book* consists of a clique, called its spine, and a set of further vertices, called its pages, each joined to every spine vertex in the same colour. We change two parts of the analysis.

The local step. The book algorithm keeps a pair of vertex sets (X, Y) with many red edges between them. At each step it either moves into the red neighbourhood of a vertex of X , or into its blue neighbourhood, or builds a blue book. Previous analyses track the red edge density between X and Y . We track instead the potential

$$\Phi_c(X, Y) = |X|^w |Y| \left(\frac{1}{|Y|} \sum_{y \in Y} (g_y - c)_+^s \right)^{r/s},$$

where g_y is the proportion of X joined in red to y , the shift c is slightly below the target density, and $w > 0$, r large, $s = \beta r$ with $0 < \beta < 1$, and $u_+ = \max\{u, 0\}$. Passing to subsets preserves

the possible clique outcomes, so we may choose a pair that maximises this potential. Maximality controls both the column densities and the contribution of each row. When few rows have large blue degree, Theorem 3.4 rules out simultaneous failure of the red and blue tests if the explicit integral $\text{CERT}(p, \mu, w, \beta, x)$ is negative. If many rows have large blue degree, we extract a blue book instead. These alternatives give the local step used in the induction.

Feedback into the off-diagonal input. The local step needs, as a stopping rule, an upper bound for $R(a, b)$ for all pairs (a, b) , not only the diagonal. Lu and Wang supply such a bound, $\log R(a, b) \leq \hat{f}(a, b) + o(a + b)$, through an explicit function $\hat{f}$. The bounds our algorithm produces are better than $\hat{f}$ in a range of ratios b/a . We therefore use these better bounds as off-diagonal inputs. Section 5 proves that a finite list of such bounds may depend on each other, even cyclically, provided every dependency is applied to a strictly smaller colouring. The proof is an induction on the number of vertices, not an assumption that the improved bounds already hold. The finite collection used for Theorem 1.1 contains all the bounds and comparisons needed to justify this induction.

Outline. Section 2 fixes notation and states the input from Lu and Wang. Section 3 proves the local step. Section 4 turns it into a bound for colourings that satisfy a stopping rule. Section 5 sets up profiles and banks and proves that a bank satisfying finitely many inequalities is valid. Section 6 describes the certified bank and completes the proof of Theorem 1.1. Section 7 describes the verification.

2 Notation and the off-diagonal input

2.1 Colourings

A *host* is a red/blue colouring of the edges of a complete graph on a finite vertex set V ; we write $|V|$ for its order. For disjoint finite sets $X, Y \subseteq V$, the red density between them is $d_R(X, Y) = e_R(X, Y)/(|X||Y|)$, where $e_R(X, Y)$ is the number of red edges with one end in X and one in Y . The red density of V itself is $d_R(V) = e_R(V)/\binom{|V|}{2}$, and $d_B(V) = 1 - d_R(V)$. For positive integers a, b , we write $V \supseteq (a, b)$ if V contains a red K_a or a blue K_b , and $N \longrightarrow (a, b)$ if every host on N vertices does. Thus $R(a, b) = \min\{N : N \longrightarrow (a, b)\}$. The property $N \longrightarrow (a, b)$ is monotone in N . We use the Erdős–Szekeres bound

$$\binom{a+b-2}{a-1} \longrightarrow (a, b).$$

Exchanging the two colours turns a host with red density d into one with red density $1 - d$, and a red K_a or blue K_b into a blue K_a or red K_b . We call this *renaming the colours*.

Lemma 2.1 (balanced cut). *Let W be a host with $|W| \geq 3$ and red density d . Then $W = X \sqcup Y$ with $d_R(X, Y) \geq d$ and $|X|, |Y| \geq |W|/3$.*

Proof. Average $d_R(X, W \setminus X)$ over all $X \subseteq W$ with $|X| = \lfloor |W|/2 \rfloor$. Every pair of distinct vertices is separated by the same number of these sets, so the average of $e_R(X, W \setminus X)$ is a fixed multiple of $e_R(W)$, and the average of $|X||W \setminus X|$ is the same multiple of $\binom{|W|}{2}$. Hence some X has $d_R(X, W \setminus X) \geq d$. Both parts have at least $\lfloor |W|/2 \rfloor \geq |W|/3$ vertices when $|W| \geq 3$. $\square$

2.2 The Lu–Wang source function

Lu and Wang [4] define a function $f : [0, 1] \rightarrow \mathbb{R}$ by

$$f(t) = (1+t) \log(1+t) - t \log t + t e^{-t} P(t) \quad (0 \log 0 = 0).$$

Here P is the piecewise cubic Hermite interpolant of 2280 rational knots $0 = t_0 < \dots < t_{2279} = 1$, with prescribed rational values and derivatives at the knots. On each interval between consecutive knots, P is the cubic polynomial matching the values and first derivatives at both endpoints. Thus P is C^1 . These data are the file `data/source.json` of the Lu–Wang repository [5] (SHA-256 `f3e4a4ea...c08c`; Appendix A). The *symmetric extension* of f is

$$\widehat{f}(\lambda) = \begin{cases} f(\lambda), & 0 < \lambda \leq 1, \\ \lambda f(1/\lambda), & \lambda \geq 1, \end{cases} \quad \widehat{f}(a, b) = a \widehat{f}(b/a) \quad (a, b > 0).$$

So $\widehat{f}(a, b) = \widehat{f}(b, a)$, and $\widehat{f}(a, b)$ is homogeneous of degree one.

Theorem 2.2 (Lu–Wang uniform source bound [4, eq. (32), Lemma B.1]). *For every $\varepsilon > 0$ there is $C_\varepsilon \geq 1$ such that, for all integers $a, b \geq 1$ and N ,*

$$N \geq C_\varepsilon \exp(\widehat{f}(a, b) + \varepsilon(a + b)) \implies N \longrightarrow (a, b).$$

Lu and Wang state and prove this in Lean as `RamseyCurrent.FiniteSourceBound`. Their theorem quantifies over all pairs, however unbalanced. Our Lean development imports their proof, not only their statement (Section 7).

Lemma 2.3 (symmetric concavity). *$\widehat{f}$ is concave on $(0, \infty)$. Consequently, for every $t > 0$ with $t \neq 1$, $\widehat{f}$ is differentiable at t and $\widehat{f}(s) \leq \widehat{f}(t) + \widehat{f}'(t)(s - t)$ for all $s > 0$. At $t = 1$ the same inequality holds with any slope in $[f(1) - f'(1), f'(1)]$.*

Proof. Lu and Wang prove that f is concave on $[0, 1]$. For $\lambda > 1$ the function $\lambda f(1/\lambda)$ is the perspective of a concave function, so it is concave on $[1, \infty)$. A function that is concave on $(0, 1]$ and on $[1, \infty)$, and continuous at 1, is concave on $(0, \infty)$ if its left derivative at 1 is at least its right derivative there. The left derivative is $f'(1)$; the right derivative is $f(1) - f'(1)$. So the condition is $2f'(1) - f(1) \geq 0$. Writing v, d for the value and derivative of P at 1, one finds $e(2f'(1) - f(1)) = 2d - v$, which is a positive rational number for the Lu–Wang data. The tangent bounds are the usual supergradient inequalities of a concave function. $\square$

The tangent bounds are what the certificate uses to compare $\widehat{f}$ with straight lines.

3 The local step

3.1 A potential that weights high-density columns

We need a quantity that measures both the sizes of the two sets and the red density seen by each vertex of Y . The moment defined below gives more weight to vertices with unusually many red neighbours in X . Maximising the resulting potential will give the row and column inequalities used in the local step.

Fix a host on V , and disjoint nonempty sets $X, Y \subseteq V$. For $v \in X$ and $y \in Y$ put $h_{vy} = 1$ if vy is red and 0 otherwise. The *column density* of y and the *row density* of v are

$$g_y = \frac{1}{|X|} \sum_{v \in X} h_{vy}, \quad \pi_v = \frac{1}{|Y|} \sum_{y \in Y} h_{vy}.$$

Both average to $d_R(X, Y)$. Fix reals $w > 0$, $0 < \beta < 1$ and $r > w$ with $s = \beta r > 1$, so $1 < s < r$. For a real shift c let

$$H_c(X, Y) = \left(\frac{1}{|Y|} \sum_{y \in Y} (g_y - c)_+^s \right)^{1/s}, \quad \Phi_c(X, Y) = |X|^w |Y| H_c(X, Y)^r,$$

with $\Phi_c = 0$ if X or Y is empty. We call (X, Y) *maximal* (for c, w, r, s) if $\Phi_c(X', Y) \leq \Phi_c(X, Y)$ for every nonempty $X' \subseteq X$ and $\Phi_c(X, Y') \leq \Phi_c(X, Y)$ for every nonempty $Y' \subseteq Y$.

The potential rewards large sets and large column densities. The moment of order s weights the columns with high red density more than an average does. The weight w sets the relative value of a vertex of X against a vertex of Y .

Throughout this section we write $N = |X|$, $M = |Y|$, $u_y = g_y - c$, $H = H_c(X, Y)$, and $\mathbb{E}_X, \mathbb{E}_Y$ for uniform averages.

Lemma 3.1 (columns at a maximal pair). *Suppose (X, Y) is maximal and $H > 0$. Then $u_y \geq (1 - \beta)^{1/s} H > 0$ for every $y \in Y$. In particular $\alpha := \mathbb{E}_Y u_y \geq (1 - \beta)^{1/s} H$.*

Proof. For nonempty $Y' \subseteq Y$ we have $\Phi_c(X, Y') = N^w |Y'|^{1-r/s} \left(\sum_{y \in Y'} (u_y)_+^s \right)^{r/s}$, and $1 - r/s < 0$. If some $u_{y_0} \leq 0$ and $M \geq 2$, deleting y_0 keeps the sum and lowers $|Y'|$, which strictly increases Φ_c ; this contradicts maximality. If $M = 1$ then $u_y = H > 0$. Now let $M \geq 2$, fix y and put $a = u_y/H$, so that $u_y^s / \sum_{y'} u_{y'}^s = a^s/M$. Maximality at $Y \setminus \{y\}$, raised to the power s/r , gives

$$1 - \frac{a^s}{M} \leq \left(1 - \frac{1}{M} \right)^{1-s/r} \leq 1 - \frac{1-\beta}{M},$$

by Bernoulli's inequality, since $1 - s/r = 1 - \beta \in (0, 1)$. Hence $a^s \geq 1 - \beta$. $\square$

Lemma 3.2 (rows at a maximal pair). *Suppose (X, Y) is maximal and $H > 0$. For $v \in X$ put*

$$V_v = \frac{\mathbb{E}_Y [u_y^{s-1} (h_{vy} - c)]}{H^s}.$$

Then $\mathbb{E}_X V_v = 1$, and for every $z \in [0, 1]^X$ with $t = \mathbb{E}_X z_v > 0$,

$$\mathbb{E}_X [z_v V_v] \leq t^{1-w/r}. \quad (1)$$

In particular $V_v \geq 1 - w/r$ for every $v \in X$.

The variable z_v allows a fractional choice of rows. Inequality (1) bounds the total contribution of such a choice by its size. The final assertion says that no row contributes much less than the average contribution, which is 1.

Proof. Averaging $h_{vy} - c$ over v gives u_y , so $\mathbb{E}_X V_v = \mathbb{E}_Y [u_y^s] / H^s = 1$. For $z \in [0, 1]^X$ let $b_y = \mathbb{E}_X [z_v (h_{vy} - c)]$ and define

$$\Lambda(z) = t \left(\frac{\| (b/t)_+ \|_s}{H} \right)^{r/w}, \quad \|v\|_s = (\mathbb{E}_Y |v_y|^s)^{1/s},$$

with $\Lambda(0) = 0$. The map $v \mapsto (\|v_+ \|_s / H)^{r/w}$ is convex, because $v \mapsto \|v_+ \|_s$ is convex and nonnegative and $r/w > 1$. So Λ is its perspective composed with a linear map, which is convex on $\{t > 0\}$. Since b/t is a convex combination of the fixed vectors $(h_{vy} - c)_y$, $\Lambda(z) = O(t)$, so Λ extends continuously by 0 and is convex on the whole cube $[0, 1]^X$. At a vertex $z = \mathbf{1}_{X'}$ with $X' \neq \emptyset$ we have

$b/t = (g'_y - c)_y$, where g' is the column density of X' , so $\Lambda(\mathbf{1}_{X'}) = (\Phi_c(X', Y)/\Phi_c(X, Y))^{1/w} \leq 1$ by maximality. A convex function on the cube is at most its largest vertex value, so $\Lambda \leq 1$, which says $\|b_+\|_s \leq H t^{1-w/r}$. By Hölder's inequality,

$$H^s \mathbb{E}_X[z_v V_v] = \mathbb{E}_Y[u_y^{s-1} b_y] \leq \|u^{s-1}\|_{s/(s-1)} \|b_+\|_s = H^{s-1} \|b_+\|_s,$$

using $u_y > 0$ (Lemma 3.1). This proves (1). Taking $z = \mathbf{1} - \epsilon e_v$ gives $1 - \epsilon V_v/N \leq (1 - \epsilon/N)^{1-w/r}$, and letting $\epsilon \downarrow 0$ gives $V_v \geq 1 - w/r$. $\square$

Remark 3.3. For $w \geq 1$ one may use convexity of Λ^w directly. For $0 < w < 1$ that function need not be convex, and taking the $1/w$ -th power is what makes the argument work. Of the 918 controls in the certified bank, 648 have $w < 1$.

3.2 An integral that rules out failure of both steps

We now define the number whose sign certifies a local step. The parameter p is the target red density, μ controls the blue step, and x is the fraction of the potential that a red step must retain. The parameters w and β are those of the potential. Fix $0 < p < 1$, $0 < \mu < 1$, $w > 0$, $0 < \beta < 1$ and $x > 0$. For $q \in (0, \mu]$ let

$$A(q) = \frac{1-q}{q} \log \frac{p(1-q)^w}{x}, \quad B(q) = w \log \frac{\mu}{q}, \quad U(q) = e^{\beta B(q)} = (\mu/q)^{w\beta}.$$

Let $M(q) = B(q)$ if $A(q) \leq B(q)$, and otherwise

$$M(q) = p A(q) + \frac{1-p}{\beta} \log \frac{U(q) - p e^{\beta A(q)}}{1-p}.$$

The logarithm in the second branch is defined only when its numerator is positive. Accordingly, the *feasible set* $\mathcal{Q} = \mathcal{Q}(p, \mu, w, \beta, x)$ is the set of $q \in (0, \mu]$ with $A(q) \leq B(q)$, or with $A(q) > B(q)$ and $p e^{\beta A(q)} < U(q)$. When $\mathcal{Q} \neq \emptyset$ we define

$$\text{CERT}(p, \mu, w, \beta, x) = \int_0^1 \sup_{q \in \mathcal{Q}} [q M(q) + w q (1 + \log t)] dt.$$

On $\mathcal{Q}$ one has $M(q) \leq B(q)$, by concavity of the logarithm, so the integrand is bounded above. The statement $\text{CERT} < 0$ includes the assertion that the integrand is integrable.

Meaning. In the application, q is the blue degree of a row of X , divided by $|X|$. Passing to its blue neighbourhood retains a fraction q^w of the factor $|X|^w$ in the potential. The quantities $A(q)$ and $B(q)$ describe the changes in the column moment needed for the red and blue steps, respectively. The function $M(q)$ bounds the average change allowed when both steps fail. Row maximality gives a second, averaged constraint after we sort rows by blue degree. Its contribution to the limiting inequality is $wq(1 + \log t)$, where t is the fraction of rows below the corresponding degree. For each t , the supremum permits any feasible degree q . If even this upper bound has negative integral, simultaneous failure is impossible. Theorem 3.4 makes this argument precise.

3.3 The sharp envelope theorem

Let $\mathcal{B}$ be any simple graph on V . For $i \in X$ let $B_i = \{j \in X : ij \in \mathcal{B}\}$, $q_i = |B_i|/N$, and let $b_i(y)$ be the proportion of B_i joined in red to y (with $b_i(y) = 0$ if $B_i = \emptyset$). In the application, $\mathcal{B}$ is the blue graph and B_i is the blue neighbourhood of i in X .

We say row i with $q_i > 0$ *fails blue* if

$$\sum_{y \in Y} (b_i(y) - c)_+^s < \left(\frac{\mu}{q_i}\right)^{w\beta} \sum_{y \in Y} (u_y)_+^s, \quad (2)$$

and row i *fails shifted-red* if

$$\sum_{y \in Y: h_{iy}=1} \left((1 - q_i)u_y - q_i(b_i(y) - g_y) \right)_+^s < (1 - q_i)^{s-w\beta} x^\beta \pi_i^{1-\beta} \sum_{y \in Y} (u_y)_+^s. \quad (3)$$

Theorem 3.4 (sharp envelope). *Let $0 < p < 1$, $0 < \mu < 1$, $w > 0$, $0 < \beta < 1$ and $0 < x < \min\{p, (1 - \mu)^w\}$. Suppose $\mathcal{Q} \neq \emptyset$ and $\text{CERT}(p, \mu, w, \beta, x) < 0$. Then there are $\eta, \tau, \zeta > 0$ and R_0 with $R_0 > w$ and $\beta R_0 > 1$ such that, for every $r \geq R_0$, there is no configuration consisting of a host on V , a simple graph $\mathcal{B}$ on V , disjoint nonempty $X, Y \subseteq V$ and a real c with $|c - p| \leq \eta$, such that*

- (i) $g_y > c$ for every $y \in Y$;
- (ii) (X, Y) is maximal for Φ_c with parameters $w, r, s = \beta r$;
- (iii) every $i \in X$ with $q_i > 0$ fails blue;
- (iv) there is $X_g \subseteq X$ with $|X \setminus X_g| \leq \zeta |X|$ such that every $i \in X_g$ has $q_i \leq \mu + \tau$ and fails shifted-red.

In words: at a maximal pair whose shift is close to p , some row of positive blue degree passes the blue test, or more than a ζ -fraction of the rows either have large blue degree or pass the shifted-red test.

The proof is by contradiction. If both tests could fail for arbitrarily large r , we would obtain a sequence of configurations with vanishing exceptional proportions. In the limit, maximality and the fact that the blue graph is undirected give an averaged inequality incompatible with $\text{CERT} < 0$. We first state this limit argument, then recover the finite statement.

Proposition 3.5 (limit form). *Fix p, μ, w, β, x as in Theorem 3.4. Suppose that for each j we have a configuration with parameter r_j satisfying (i)–(iv) of Theorem 3.4 with $\tau = \tau_j$ and $\zeta = \zeta_j$, where $r_j \rightarrow \infty$, $r_j > w$, $\beta r_j > 1$, $\tau_j \rightarrow 0$, $\zeta_j \rightarrow 0$, the shifts c_j and the moments H_j are bounded, and $\liminf_j d_j \geq p$ where $d_j = d_R(X_j, Y_j)$. Then, if $\mathcal{Q} \neq \emptyset$ and the integrand of CERT is integrable, $\text{CERT}(p, \mu, w, \beta, x) \geq 0$.*

Proof of Theorem 3.4 from Proposition 3.5. Suppose the conclusion fails. Then for every $j \geq 1$ the choice $\eta = \tau = \zeta = 1/j$ and $R_0 = \max\{j, w + 1, 2/\beta\}$ admits some $r_j \geq R_0$ and a configuration with $|c_j - p| \leq 1/j$ satisfying (i)–(iv). Then $|c_j| \leq 1 + p$, $H_j \leq \max_y u_y \leq 1 - c_j \leq 2$, and $d_j = c_j + \mathbb{E}_Y u_y > c_j \geq p - 1/j$. So Proposition 3.5 applies and gives $\text{CERT} \geq 0$, a contradiction. $\square$

The proof has four main parts. We first bound what one row can contribute when both tests fail. We then show that almost all rows have the same limiting red density. Next we use row maximality and the symmetry of the blue graph to bound the average contribution from below. Finally we pass to the limit and compare the result with the integral defining CERT . We give all eight steps below.

The rest of this subsection proves Proposition 3.5. Constants hidden in $O(\cdot)$ depend only on β , w and the bounds on $|c|$ and H . Fix one configuration. Since all $u_y > 0$, put $a_y = u_y/H$, so $\mathbb{E}_Y a_y^s = 1$, and let ν be the probability measure on Y with $\nu_y = a_y^s/M$. Put $m_i = \mathbb{E}_\nu h_{iy}$ and, for $q_i > 0$, $Z_i(y) = (b_i(y) - g_y)/u_y$. Since $u_y + (b_i(y) - g_y) = b_i(y) - c$,

$$\mathbb{E}_\nu(1 + Z_i)_+^s = \frac{\sum_y (b_i(y) - c)_+^s}{\sum_y u_y^s},$$

$$\mathbb{E}_\nu \left[h_{iy} ((1 - q_i) - q_i Z_i)_+^s \right] = \frac{\sum_{y: h_{iy}=1} ((1 - q_i)u_y - q_i(b_i(y) - g_y))_+^s}{\sum_y u_y^s}.$$

So (2) says $\mathbb{E}_\nu(1 + Z_i)_+^s < U_i := (\mu/q_i)^{w\beta}$, and (3) says $\mathbb{E}_\nu[h_{iy}((1 - q_i) - q_i Z_i)_+^s] < (1 - q_i)^{s-w\beta} x^\beta \pi_i^{1-\beta}$. For $q_i = 0$ the left side of (3) is m_i .

Step 1: an optimisation on two parts of Y . For a fixed row, split Y according to whether its edge to the row vertex is red. Jensen's inequality reduces the moment constraints to one number on each part. Let $0 < m < 1$, $s > 1$, $U > 0$ and $t_U = U^{1/s}$. Consider the supremum of $mz_{\text{in}} + (1 - m)z_{\text{out}}$ subject to $z_{\text{in}} \geq L$ and $m(1 + z_{\text{in}})_+^s + (1 - m)(1 + z_{\text{out}})_+^s < U$. Raising a coordinate below -1 to -1 keeps the constraint and raises the objective, so we may write $\mathcal{A} = 1 + z_{\text{in}} \geq 0$ and $\mathcal{C} = 1 + z_{\text{out}} \geq 0$. By Jensen, $m\mathcal{A} + (1 - m)\mathcal{C} < t_U$. If $L \leq t_U - 1$, the supremum is $t_U - 1$. The constraint is strict, so these boundary values are suprema, not maxima. If $L > t_U - 1$, then on the boundary of the closed constraint $\mathcal{C} = ((U - m\mathcal{A}^s)/(1 - m))^{1/s}$ the objective has derivative $m[1 - (\mathcal{A}/\mathcal{C})^{s-1}] < 0$ in $\mathcal{A}$. Thus, when the strict constraint is feasible, the supremum is approached with $\mathcal{A} = 1 + L$ and $\mathcal{C}$ tending to its boundary value from below. It equals

$$mL + (1 - m) \left[\left(\frac{U - m(1 + L)^s}{1 - m} \right)^{1/s} - 1 \right], \quad (4)$$

The feasible set in this branch is nonempty exactly when $U > m(1 + L)^s$.

For a row $i \in X_g$ with $q_i, m_i > 0$, Jensen's inequality on the red cell $\{h_{iy} = 1\}$ (of ν -mass m_i), applied to (3), gives $m_i((1 - q_i) - q_i z_{\text{in}})_+^s < (1 - q_i)^{s-w\beta} x^\beta \pi_i^{1-\beta}$, where $z_{\text{in}} = \mathbb{E}_\nu[Z_i \mid h_{iy} = 1]$. Taking s -th roots, and using $(s - w\beta)/s = 1 - w/r$ and $\beta/s = 1/r$,

$$z_{\text{in}} > L_i := \frac{(1 - q_i) - (1 - q_i)^{1-w/r} x^{1/r} (\pi_i^{1-\beta}/m_i)^{1/s}}{q_i}, \quad q_i r L_i = (1 - q_i) r (1 - e^{\ell_i/r}), \quad (5)$$

with $\ell_i = \log x - w \log(1 - q_i) + \beta^{-1}[(1 - \beta) \log \pi_i - \log m_i]$. If $\pi_i, m_i \rightarrow d$ and $q_i \rightarrow q$, then $r L_i \rightarrow A_d(q)$, where A_d is A with p replaced by d . Jensen on both cells, applied to (2), gives the constraint of the two-cell problem with $U = U_i$, with z_{in} as above and $z_{\text{out}} = \mathbb{E}_\nu[Z_i \mid h_{iy} = 0]$; its objective is $\mathbb{E}_\nu Z_i$.

Step 2: rows and columns concentrate. Lemma 3.1 gives $a_y^s \geq 1 - \beta$. With $K_\beta = -\log(1 - \beta)/\beta$ and $\alpha = \mathbb{E}_Y u_y$, this gives $e^{-K_\beta/r} \leq \alpha/H \leq 1$, where the upper bound is the power-mean inequality. Maximality under restricting Y to a subset A of proportion θ gives $\mathbb{E}_Y[a_y^s \mathbf{1}_A] \leq \theta^{1-s/r}$; applied to $A = \{a_y \geq t\}$ it gives $\mathbb{P}_Y(a_y \geq t) \leq t^{-r}$ for $t \geq 1$. Let $J = \mathbb{E}_Y a^{s-1} \leq 1$ and $P_+ = \mathbb{E}[(a - 1)_+ a^{s-1}]$. The layer-cake formula with the tail bound gives

$$P_+ \leq \int_1^\infty (st^{s-1} - (s - 1)t^{s-2})t^{-r} dt = \frac{r}{(r - s)(r - s + 1)} = O(1/r),$$

and $1 - J = \mathbb{E}[a^{s-1}(a-1)] \leq P_+$.

Let $f_i = \mathbb{E}_Y[a^{s-1}h_{iy}] = HV_i + cJ$. Then $\mathbb{E}_X f = H + cJ$. By Lemma 3.2, $f_i - \mathbb{E}f \geq -Hw/r$, and since $f - \mathbb{E}f$ has mean zero, $\mathbb{E}_X|f_i - \mathbb{E}f| \leq 2Hw/r$. Also $m_i = \mathbb{E}_Y[a^s h_{iy}]$, so $|m_i - f_i| \leq \mathbb{E}_Y[a^{s-1}|a-1|] = 2P_+ - (1-J) \leq 2P_+$. Finally $\mathbb{E}f - d = (H - \alpha) - c(1-J)$. Together,

$$\mathbb{E}_X|m_i - d| = O(1/r). \quad (6)$$

For the ordinary row densities, maximality and the power-mean inequality give $\mathbb{P}_X(\pi_i \geq c + Ht) \leq t^{-r/w}$ for $t > 1$: the set A of such rows has $H_c(A, Y) \geq Ht$, so $(|A|/N)^w t^r \leq 1$. Integrating, $\mathbb{E}(\pi - c - H)_+ \leq Hw/(r-w)$, and since $\mathbb{E}\pi = c + \alpha$,

$$\mathbb{E}_X|\pi_i - d| \leq \frac{2Hw}{r-w} + 2(H - \alpha) = O(1/r). \quad (7)$$

Step 3: the averaged constraint from row maximality. We sort rows by blue degree and apply (1) to initial portions of this ordering. Let $\gamma = 1 - w/r$ and let Q be the increasing quantile function of the law of q_i under uniform $i \in X$, with distribution function $F(z) = \mathbb{P}(q \leq z)$. Thus $Q(t)$ is the blue degree at proportion t in the ordered list, with the usual quantile interpretation at atoms. Integrating a function of $Q(t)$ over $0 < t < 1$ is the same as averaging that function of q_i over the rows. The ordering lets us use row maximality to control every initial portion of the list, not just one row. By the layer-cake formula and (1) applied to $z_i = \mathbf{1}[q_i \leq z]$,

$$\mathbb{E}[qV] = 1 - \int_0^1 \mathbb{E}[V \mathbf{1}_{q \leq z}] dz \geq 1 - \int_0^1 F(z)^\gamma dz = \int_0^1 Q(t) \gamma t^{\gamma-1} dt.$$

Subtracting $\mathbb{E}q = \int_0^1 Q$,

$$r \mathbb{E}[q(V-1)] \geq \int_0^1 Q(t) k_r(t) dt, \quad k_r(t) = r[(1-w/r)t^{-w/r} - 1] \longrightarrow -w(1 + \log t), \quad (8)$$

with convergence in $L^1(0,1)$ (dominated by $C_w t^{-1/2}(1 + |\log t|)$ for $r \geq 2w$).

Step 4: symmetry of the blue graph. Averaging over blue neighbours can be done in either order, because $\mathcal{B}$ is undirected: $j \in B_i$ if and only if $i \in B_j$. Hence, for each y , $\mathbb{E}_i[q_i b_i(y)] = N^{-2} \sum_i \sum_{j \in B_i} h_{jy} = \mathbb{E}_j[q_j h_{jy}]$. Since $\sum_y \nu_y(h_{iy} - g_y)/u_y = V_i - 1$, this gives

$$\mathbb{E}_X[q_i \mathbb{E}_\nu Z_i] = \mathbb{E}_X[q_i(V_i - 1)]. \quad (9)$$

Both (8) and (9) hold over all rows, including the exceptional ones.

Step 5: small blue degrees do not occur. Call a row *regular* if $|\pi_i - d|$ and $|m_i - d|$ are at most η_j , where $\eta_j \rightarrow 0$ slowly enough that, by (6), (7) and Markov's inequality, the proportion of non-regular rows tends to 0. Let $\kappa = \log(p/x) > 0$. For regular rows in X_g with $0 < q \leq \delta_0$ and j large, $\ell \leq -\kappa/2$ and $m \geq p/2$, so (5) gives $q r L \geq \kappa/8$. On the other hand, the blue constraint on the red cell gives $m(1 + z_{\text{in}})^s < U$, so

$$q r z_{\text{in}} < q r (e^{b/r} - 1) \leq q b e^{b/r}, \quad b = w \log(\mu/q) - \beta^{-1} \log m,$$

and the right side is at most $C\sqrt{q}(1 + \log(1/q))$ for $r \geq 2w$, $m \geq p/2$ and $q \leq \mu$. Since $z_{\text{in}} > L$, rows with $0 < q < \delta$ cannot occur, for a fixed small $\delta > 0$. At $q = 0$, (3) reads $m < x^\beta \pi^{1-\beta}$, which

is impossible when π and m are close to $d \geq p - o(1) > x$. Hence every weak limit of the laws of q_i is supported on $[\delta, \mu]$; the upper end comes from $q_i \leq \mu + \tau_j$ on X_g . Such weakly convergent subsequences exist because these are probability measures on the compact interval $[0, 1]$. The exceptional rows have vanishing proportion, so they contribute no mass outside $[\delta, \mu]$ in the limit. This support restriction keeps the logarithms in the limiting argument away from $q = 0$.

Step 6: the limiting envelope. Pass to a subsequence with $d_j \rightarrow d_* \in [p, 1]$. The estimates in Step 2 make both π_i and m_i converge in probability to this same density. The remaining variable is the blue degree q . We now find the largest scaled row gain compatible with a given limiting value of q . Combining $z_{\text{in}} > L$ with the blue constraint gives $m(1 + L)_+^s < U$. At a regular limit point $q \in [\delta, \mu]$ we get $rL \rightarrow A_{d_*}(q)$ and $(1 + L)^s \rightarrow e^{\beta A_{d_*}(q)}$, so

$$d_* e^{\beta A_{d_*}(q)} \leq U(q). \quad (10)$$

The inequality is non-strict because a strict inequality can become equality in a limit. In logarithms, (10) says $T_{d_*}(q) \leq \log x$, where $T_d(q) = \left(1 + \frac{q}{\beta(1-q)}\right) \log d + w \log(1 - q) + \frac{wq}{1-q} \log \frac{q}{\mu}$. Since $T'_d(q) = [\beta^{-1} \log d + w \log(q/\mu)]/(1 - q)^2 \leq 0$ for $d \leq 1$ and $q \leq \mu$, we get $T_{d_*}(\mu) \leq \log x$, that is,

$$d_* \leq [x/(1 - \mu)^w]^{1/(1 + \mu/(\beta(1 - \mu)))} < 1,$$

using $x < (1 - \mu)^w$. For q in a compact subset of $(0, 1)$ and $\pi, m \rightarrow d \in (0, 1)$, r times the two-cell supremum (4) converges to $M_d(q)$, where M_d is M with p replaced by d . In the non-binding case $r(t_U - 1) \rightarrow w \log(\mu/q) = B(q)$. In the binding case $m r L \rightarrow d A_d$ and $r[(U - m(1 + L)^s)/(1 - m)]^{1/s} - 1 \rightarrow \beta^{-1} \log \frac{U - d e^{\beta A_d}}{1 - d}$. The two branches agree when $A_d = B$. At the boundary $U = d e^{\beta A_d}$ the scaled outside-cell term tends to $-\infty$, and outside the feasible set $\mathcal{Q}_d$ (defined with d in place of p) there are eventually no feasible parameters. So the limit is upper semicontinuous with values in $[-\infty, \infty)$, where we set $M_d = -\infty$ off $\mathcal{Q}_d$.

Step 7: passage to the limit. Jensen applied to (2) gives $(1 + \mathbb{E}_\nu Z)^s < U$, so for $q > 0$,

$$q r \mathbb{E}_\nu Z \leq q r [(\mu/q)^{w/r} - 1],$$

which is at most $w \mu^{w/r} q^{1-w/r} \log(\mu/q)$ for $q \leq \mu$, uniformly bounded for $r \geq 2w$, and is ≤ 0 for $q > \mu$. Rows with $q = 0$ contribute 0. So $q r \mathbb{E}_\nu Z$ has a common upper bound H_0 over all rows. On regular rows in X_g it is also at most q times r times the two-cell supremum. Define a dominating random variable W_j by taking the smaller of these two bounds on those rows, and H_0 on the exceptional rows. The latter have vanishing proportion.

Take a subsequence realising the limsup and then a weakly convergent subsequence of the joint laws of (q, m, π, ι) , where ι indicates that the row is regular and lies in X_g . These laws live on a compact metric space. Skorokhod's theorem represents them on one probability space with almost-sure convergence, while preserving their distributions and hence their expectations. The limits have $m = \pi = d_*$ and $\iota = 1$. In particular, almost every coupled row is non-exceptional for all sufficiently large j , so Step 6 bounds the pointwise upper limit of W_j by $q M_{d_*}(q)$. Applying Fatou's lemma to the nonnegative variables $H_0 - W_j$ now gives an upper bound on the limiting expectation. Thus

$$\limsup_j \mathbb{E}_X [q r \mathbb{E}_\nu Z] \leq \int_0^1 Q(t) M_{d_*}(Q(t)) dt,$$

where Q is the quantile of the limit law of q . This integral is simply the expectation of the limiting envelope, written in terms of quantiles.

For the lower bound we retain the averaged information from row maximality and blue-graph symmetry. Equations (9) and (8) give $\mathbb{E}_X[q r \mathbb{E}_\nu Z] = r \mathbb{E}[q(V-1)] \geq \int_0^1 Q_j k_{r_j}$. The quantiles Q_j converge to Q at every continuity point of Q and are bounded by 1, and $k_{r_j} \rightarrow -w(1 + \log t)$ in L^1 . The monotone function Q has at most countably many discontinuities. Dominated convergence handles the product of $Q_j - Q$ with the limiting integrable kernel, while the L^1 convergence of the kernels handles the remaining error. Therefore

$$\liminf_j \mathbb{E}_X[q r \mathbb{E}_\nu Z] \geq -w \int_0^1 Q(t)(1 + \log t) dt.$$

Together,

$$\int_0^1 [Q(t) M_{d_*}(Q(t)) + w Q(t)(1 + \log t)] dt \geq 0, \quad (11)$$

and in particular the integral is finite. This is the constraint that will contradict a negative certificate: the limiting degree distribution must satisfy it, even though no contradiction need hold for an individual row.

Step 8: from the limit to CERT. Fix $q \in (0, \mu]$. On the binding branch put $E = e^{\beta A}$ and $z = (U - dE)/(1 - d)$. There $E > U > z > 0$. Differentiating $M = dA + \frac{1-d}{\beta} \log z$ gives $\beta \partial_d M = \log(E/z) + 1 - E/z \leq 0$ at fixed A and $\partial_A M = d(1 - E/z) < 0$, while $\partial_d A_d = (1 - q)/(qd) > 0$. So $M_d(q)$ decreases in d . On the non-binding branch $M_d = B$, and the branches meet continuously. Also $de^{\beta A_d(q)}$ increases in d , so $\mathcal{Q}_d$ shrinks as d grows. Since $d_* \geq p$, we get $\mathcal{Q}_{d_*} \subseteq \mathcal{Q}$ and $M_{d_*} \leq M$ on $\mathcal{Q}_{d_*}$. Since (11) is finite, $Q(t) \in \mathcal{Q}_{d_*}$ for almost every t . For such t ,

$$Q(t) M_{d_*}(Q(t)) + w Q(t)(1 + \log t) \leq \sup_{q \in \mathcal{Q}} [q M(q) + w q(1 + \log t)].$$

Integrating gives $\text{CERT} \geq 0$. This proves Proposition 3.5, and hence Theorem 3.4. $\square$

Remark 3.6. Two points of the proof deserve emphasis. A single row can fail both tests at the parameters we use; the contradiction comes only from the averaged inequality (11), that is, from the drift (8) together with stationarity (9). And Step 5 is essential: allowing $q = 0$ in the supremum would make the integrand nonnegative and destroy every certificate.

4 Closing the induction inside one host

This section turns the local step into a clique outcome for a pair (X, Y) inside a host. We assume an off-diagonal bound on proper subsets, so that we can stop when a subset is already large enough to contain one of the required cliques. Otherwise we pass to a red neighbourhood, a blue neighbourhood, or the pages of a blue book. Each move reduces a clique target and retains enough potential to continue. The induction follows Lu and Wang [4, Lemma A.1], with their density potential replaced by Φ_c .

Definition 4.1 (stopping rule). For reals $C \geq 1$, $\hat{x}, \hat{y} \in (0, 1)$ and an integer $\ell \geq 1$, a host on V satisfies $\text{Stop}(C, \hat{x}, \hat{y}, \ell)$ if, for every integer $a \geq 1$, every proper subset $Z \subsetneq V$ with $|Z| \geq C \hat{x}^{-a} \hat{y}^{-\ell}$ contains a red K_a or a blue K_ℓ .

Theorem 4.2 (finite-host closure). Let $0 < p < 1$, $0 < \mu < 1$, $w > 0$, $0 < \beta < 1$ and $0 < x < \min\{p, (1 - \mu)^w\}$, and suppose $\mathcal{Q} \neq \emptyset$ and $\text{CERT}(p, \mu, w, \beta, x) < 0$. Let $\xi, y_0, \nu, \hat{x}, \hat{y} \in (0, 1)$ satisfy

$$\xi < \min\{x, \hat{x}\}, \quad y_0 < \hat{y}, \quad \nu < \mu.$$

Then there is an integer L_0 , depending only on these ten numbers, with the following property. Let $C \geq 1$, let $\ell \geq L_0$, and let a host on V satisfy $\text{Stop}(C, \hat{x}, \hat{y}, \ell)$. Let $k, t \geq 1$ be integers and $X, Y \subseteq V$ disjoint and nonempty with

$$d_R(X, Y) \geq p, \quad |X|^w |Y| \geq C^{w+1} \xi^{-k} y_0^{-\ell} \nu^{-wt}.$$

Then $X \cup Y$ contains a red K_k , or X contains a blue K_t , or Y contains a blue K_ℓ .

The order of quantifiers matters later: L_0 is chosen before C and before the host.

4.1 Two lemmas on books

Lemma 4.3 (good pages). *Let (X, Y) be maximal with $H = H_c(X, Y) > 0$, and let*

$$D_r = \frac{w}{r-w} + 1 - (1-\beta)^{1/(\beta r)}, \quad \mathcal{G} = \{v \in X : \pi_v \geq c\}.$$

Then $1 - |\mathcal{G}|/|X| \leq D_r$.

Proof. For $t > 1$, the rows with $\pi_v \geq c + Ht$ form a set A whose column densities average at least $c + Ht$, so $H_c(A, Y) \geq Ht$ by the power-mean inequality. Maximality gives $(|A|/N)^w t^r \leq 1$, so $\mathbb{P}_X(\pi_v \geq c + Ht) \leq t^{-r/w}$. The layer-cake formula gives $\mathbb{E}_X(\pi_v - c - H)_+ \leq Hw/(r-w)$. Since $\mathbb{E}_X \pi_v = c + \alpha$ and $\alpha \geq (1-\beta)^{1/s} H$ (Lemma 3.1),

$$\mathbb{E}_X(c + H - \pi_v)_+ = \mathbb{E}_X(\pi_v - c - H)_+ - (\alpha - H) \leq D_r H.$$

A row outside $\mathcal{G}$ has $(c + H - \pi_v)_+ > H$, so Markov's inequality gives the claim. $\square$

Lemma 4.4 (blue book with good pages). *Let X be a nonempty vertex set of a host, $N = |X|$, and $\mathcal{G} \subseteq X$. Let $0 < \theta < \tau < 1$, D real, and $b \leq m$ positive integers with*

$$1 - \frac{|\mathcal{G}|}{N} \leq D \leq \frac{1}{4}, \quad \frac{m}{N} \leq \frac{1}{4}, \quad D + \frac{m}{N} + \frac{b}{m} \leq \tau - \theta.$$

Let W be the set of $v \in X$ with at least τN blue neighbours in X . If $|W| \geq R(k, m)$, then X contains a red K_k , or there are disjoint $S, T \subseteq X$ such that S is a blue clique of exactly b vertices, all S - T edges are blue, $T \subseteq \mathcal{G}$, and $|T| \geq \theta^b N/2$.

Proof. If W has no red K_k , it has a blue clique U with $|U| = m$. Let $P = \mathcal{G} \setminus U$; then $|P| \geq (1-D)N - m \geq N/2$. Each $u \in U$ has at least $(\tau - D)N - m$ blue neighbours in P , so the blue density σ between U and P satisfies $\sigma \geq \tau - D - m/N \geq \theta + b/m$. Choose a b -subset $S \subseteq U$ uniformly at random, and let T be the set of $v \in P$ joined in blue to all of S . If v has j_v blue neighbours in U , then $\mathbb{P}(v \in T) = \binom{j_v}{b} / \binom{m}{b}$. The piecewise-linear interpolation of $j \mapsto \binom{j}{b}$ is nondecreasing and convex, so by Jensen's inequality

$$\mathbb{E}|T| \geq |P| \frac{\binom{\lfloor \sigma m \rfloor}{b}}{\binom{m}{b}} \geq |P| \left(\sigma - \frac{b}{m} \right)^b \geq \frac{N}{2} \theta^b,$$

since each factor $(\lfloor \sigma m \rfloor - i)/(m - i)$ with $0 \leq i < b$ is at least $\sigma - b/m > 0$. Some choice of S achieves the expectation. $\square$

4.2 The local alternative

For $v \in X$ write B_v for the blue neighbourhood of v in X , $q_v = |B_v|/|X|$, R_v for its red neighbourhood in X , and Y_v for its red neighbourhood in Y .

Lemma 4.5. *Fix an integer N_0 with $N_0(1 - \mu) > 1$. Under the hypotheses of Theorem 4.2 on p, μ, w, β, x , there are $\eta, \tau, \zeta > 0$ and R_0 with $\eta < \min\{p, 1 - p\}$, $\text{cap} := \mu + \tau < 1 - 1/N_0$, $R_0 > w$ and $\beta R_0 > 1$, such that the following holds for every $r \geq R_0$. Let (X, Y) be maximal for Φ_c with $|c - p| \leq \eta$, $N = |X| \geq N_0$, $H > 0$, and at most ζN rows with $q_v > \text{cap}$. Then for every $\Delta \geq (1 - c)/(N(1 - \text{cap}) - 1)$ and $c' = c - \Delta$,*

(RED) *some v with $q_v \leq \text{cap}$ has $\Phi_{c'}(R_v, Y_v) \geq x \Phi_c(X, Y)$, or*

(BLUE) *some v has $\Phi_c(B_v, Y) \geq \mu^w \Phi_c(X, Y)$.*

Proof. Take η, τ, ζ, R_0 from Theorem 3.4 and shrink η, τ as required; shrinking keeps the theorem true. Suppose both alternatives fail. Apply Theorem 3.4 with $\mathcal{B}$ the blue graph. Its condition (i) is Lemma 3.1. Failure of (BLUE) at v with $q_v > 0$ reads $q_v^w H_c(B_v, Y)^r < \mu^w H^r$; raising to the power β gives (2).

Now let $X_g = \{v : q_v \leq \text{cap}\}$ and fix $v \in X_g$. Put $\sigma_v = |R_v|/N = 1 - 1/N - q_v > 0$, and let ρ_y be the red density from R_v to y . Counting red neighbours of $y \in Y_v$ in $X = \{v\} \cup B_v \cup R_v$ gives $N g_y = 1 + q_v N b_v(y) + \sigma_v N \rho_y$, so

$$\sigma_v(\rho_y - c') = (1 - q_v)u_y - q_v(b_v(y) - g_y) + \left[\sigma_v \Delta - \frac{1 - c}{N} \right],$$

and the bracket is nonnegative by the choice of Δ . With $a_y = u_y/H$ and $J = \mathbb{E}_Y a_y^{s-1} > 0$ we have $\mathbb{E}_Y [h_{vy} a_y^{s-1}] = H V_v + c J$, which is positive because $V_v \geq 1 - w/r > 0$ (Lemma 3.2) and $c \geq p - \eta > 0$. So $Y_v \neq \emptyset$. Failure of (RED) at v , raised to the power β , gives

$$\sum_{y \in Y_v} (\rho_y - c')_+^s < x^\beta \sigma_v^{-w\beta} \pi_v^{1-\beta} \sum_{y \in Y} u_y^s.$$

Using the identity and $\sigma_v \leq 1 - q_v$, this implies (3) at v . So all hypotheses of Theorem 3.4 hold, a contradiction. $\square$

4.3 Proof of Theorem 4.2

Parameters. Choose λ, y with $\xi < \lambda < \min\{x, \hat{x}\}$ and $y_0 < y < \hat{y}$. Take $N_0, \eta, \tau, \zeta, R_0, \text{cap}$ from Lemma 4.5, and θ with $\mu < \theta < \text{cap}$. Fix $r \geq R_0$ so large that $D_r < \min\{1/4, (\text{cap} - \theta)/2\}$. Fix $0 < \delta < \min\{p/2, \eta\}$, and for $n \geq 2$ let $c_n = p - \delta/n$ and $\Phi_n = \Phi_{c_n}$. Put

$$\mathcal{A} = \min\{\lambda/\xi, (\mu/\nu)^w\} > 1, \quad h = \text{cap} - \theta - D_r > 0, \quad a_0 = \min\left\{\frac{1}{w} \log \frac{\hat{x}}{\lambda}, \frac{1}{w} \log \frac{\hat{y}}{y}, \log \frac{1}{\mu}\right\} > 0,$$

and, for $n \geq 4$,

$$b_n = \max\left\{1, \left\lceil \frac{r \log(n^2/\delta) + w \log 2}{w \log(\theta/\mu)} \right\rceil\right\}, \quad m_n = \left\lceil \frac{2b_n}{h} \right\rceil.$$

Both are $O(\log n)$, and $b_n \leq m_n$. Let $\kappa_0 = \inf_{n \geq 2} \mathcal{A}^n/n^r > 0$. For $2 \leq k \leq n$ we have $R(k, m) \leq n^{m-1}$ by Erdős–Szekeres, and $n^{m_n-1} = e^{O((\log n)^2)}$, so

$$K_1 = \sup_{n \geq 4} e^{-a_0 n} \max\left\{N_0, 4m_n, \frac{2m_n}{h}, \frac{n^{m_n-1}}{\zeta}, \frac{1 + n(n-1)/\delta}{1 - \text{cap}}\right\} < \infty.$$

Let L_0 be an integer with $(y/y_0)^{L_0} \delta^r \kappa_0 \geq 1$ and $e^{a_0 L_0} \geq K_1$.

The induction. Fix $C \geq 1$, $\ell \geq L_0$ and a host satisfying $\text{Stop}(C, \hat{x}, \hat{y}, \ell)$. We prove by strong induction on $n = k + t$:

(I_n) every disjoint nonempty pair X, Y with $\Phi_n(X, Y) \geq C^{w+1} \lambda^{-k} y^{-\ell} \mu^{-wt}$ has an *outcome*: a red K_k in $X \cup Y$, a blue K_t in X , or a blue K_ℓ in Y .

If $k = 1$ or $t = 1$, any vertex of X is an outcome.

The theorem follows from (I_n). Given X, Y as in the theorem, $H_{c_n}(X, Y) \geq d_R(X, Y) - c_n \geq \delta/n$ by the power-mean inequality. Hence

$$\frac{\Phi_n(X, Y)}{C^{w+1} \lambda^{-k} y^{-\ell} \mu^{-wt}} \geq \left(\frac{\delta}{n}\right)^r \left(\frac{\lambda}{\xi}\right)^k \left(\frac{y}{y_0}\right)^\ell \left(\frac{\mu}{\nu}\right)^{wt} \geq \left(\frac{y}{y_0}\right)^\ell \delta^r \frac{\mathcal{A}^n}{n^r} \geq 1.$$

Inductive step ($k, t \geq 2$, so $n \geq 4$). Among all pairs of nonempty subsets of X and Y , choose one that maximises Φ_n . It is maximal, its potential is at least $\Phi_n(X, Y)$, and an outcome for it is an outcome for (X, Y) ; rename it (X, Y) . Write $N = |X|$, $M = |Y|$ and $H = H_{c_n}(X, Y) \in (0, 1]$, so $\Phi_n \leq N^w M$.

If $M \geq C \hat{x}^{-k} \hat{y}^{-\ell}$, then Y is a proper subset of V , and the stopping rule gives a red K_k or a blue K_ℓ in Y . Otherwise $N^w M \geq C^{w+1} \lambda^{-k} y^{-\ell} \mu^{-wt}$ and $M < C \hat{x}^{-k} \hat{y}^{-\ell}$ give

$$N > C \left(\frac{\hat{x}}{\lambda}\right)^{k/w} \left(\frac{\hat{y}}{y}\right)^{\ell/w} \mu^{-t} \geq e^{a_0(n+\ell)} \geq e^{a_0 n} K_1.$$

So $N \geq N_0$, $N \geq 4m_n$, $N \geq 2m_n/h$, $R(k, m_n) \leq \zeta N$ and $N(1 - \text{cap}) - 1 \geq n(n-1)/\delta$.

Book case: at least $R(k, m_n)$ rows have $q_v \geq \text{cap}$. Lemma 4.3 gives $1 - |\mathcal{G}|/N \leq D_r$ for $\mathcal{G} = \{v : \pi_v \geq c_n\}$. Apply Lemma 4.4 with $\tau = \text{cap}$, $b = b_n$, $m = m_n$; its conditions hold since $D_r + m_n/N + b_n/m_n \leq D_r + h = \text{cap} - \theta$. If it gives a red K_k we are done. Otherwise it gives a blue b_n -clique S and a set $T \subseteq \mathcal{G}$ of at least $\theta^{b_n} N/2$ pages. If $b_n \geq t$, then S contains a blue K_t . If $b = b_n < t$, the columns of T have average red density at least c_n , so $H_{c_{n-b}}(T, Y) \geq c_n - c_{n-b} \geq \delta/n^2$, and

$$\frac{\Phi_{n-b}(T, Y)}{\Phi_n(X, Y)} \geq 2^{-w} \theta^{wb} (\delta/n^2)^r \geq \mu^{wb}$$

by the choice of b_n . So (I_{n-b}) applies to (T, Y) with targets $(k, t-b)$. A blue K_{t-b} in T together with S is a blue K_t in X ; the other outcomes are outcomes.

Local case: fewer than $R(k, m_n) \leq \zeta N$ rows have $q_v \geq \text{cap}$. Apply Lemma 4.5 with $c = c_n$ and $\Delta = c_n - c_{n-1} = \delta/(n(n-1))$, which is large enough. In case (RED), $\Phi_{n-1}(R_v, Y_v) \geq x \Phi_n(X, Y) \geq C^{w+1} \lambda^{-(k-1)} y^{-\ell} \mu^{-wt}$, since $x > \lambda$. By (I_{n-1}) with targets $(k-1, t)$, a red K_{k-1} in $R_v \cup Y_v$ together with v is a red K_k ; the other outcomes are outcomes. In case (BLUE), lowering the shift to c_{n-1} can only increase H , so $\Phi_{n-1}(B_v, Y) \geq \mu^w \Phi_n(X, Y) \geq C^{w+1} \lambda^{-k} y^{-\ell} \mu^{-w(t-1)}$. By (I_{n-1}) with targets $(k, t-1)$, a blue K_{t-1} in B_v together with v is a blue K_t in X . $\square$

5 Profiles and banks

Theorem 4.2 needs a stopping rule, that is, an off-diagonal Ramsey bound on proper subsets of the host. This section shows how a finite collection of bounds can supply each other's stopping rules. We call this collection a *bank*. It contains bounds without a density restriction, called source pairs, and bounds for hosts of a specified density, called profiles. Controls specify the parameters with which the local step converts a source bound into a profile.

A two-bound example. Suppose we have two density-dependent bounds, with a common constant $C \geq 1$. The first says that a host of red density at least $1/2$ contains a red K_k or a blue K_b when $|V| \geq Ce^{kL_R(b/k)}$. The second has the same form with a function L_B , and can be applied after interchanging the colours and the targets. We use both bounds only at ratios in their respective domains.

Put $r = b/k$. Every host has red density at least $1/2$ or blue density at least $1/2$. In the first case the required threshold is $Ce^{kL_R(r)}$. In the second, interchanging the colours changes the targets to (b, k) , so the threshold is

$$Ce^{bL_B(k/b)} = Ce^{krL_B(1/r)}.$$

Thus $|V| \geq Ce^{k \max(L_R(r), rL_B(1/r))}$ guarantees a red K_k or a blue K_b without any density assumption. For example, at $b = 2k$ the two exponents are $kL_R(2)$ and $2kL_B(1/2)$. If $L_R(r) = a_R + b_Rr$ and $L_B(s) = a_B + b_Bs$ are positive affine functions on the intervals under consideration, then the exponent per k is simply

$$\max\{a_R + b_Rr, b_B + a_Br\}.$$

A line $A + Br$ with $A, B > 0$ lying above this maximum supplies a source bound on that interval. In the full construction, separate comparisons with the Lu-Wang input handle the remaining ratios. This example explains both the reciprocal ratio in a colour reversal and the finite line comparisons below.

Why feedback is not circular. A source bound is used by the local step only on a proper subset of the current host. Assume all bank claims have been proved for hosts with fewer vertices than V . We first obtain the profiles that use the local step on V . Next we combine these profiles on V , as in the example, and finally obtain the source bounds on V . Consequently a source bound may help prove a profile which later helps prove that same source bound: the first use is on a smaller host. The full argument below makes this induction simultaneous for all claims and chooses one constant C for the finite collection.

5.1 Controls and profiles

Definition 5.1 (source pairs). A *bank* has finitely many *source pairs* (A_h, B_h) with $A_h, B_h > 0$. Each pair stands for the claim

$$(\text{SOURCE}_h) \quad |V| \geq Ce^{A_h k + B_h b} \implies V \supseteq (k, b) \quad \text{for all integers } k, b \geq 1.$$

We write $T_h(\lambda) = A_h + B_h \lambda$, so that $A_h k + B_h b = k T_h(b/k)$.

Definition 5.2 (controls). A *control* is a tuple $\gamma = (p, \mu, w, \beta, x; \sigma; \xi, y_0, \nu)$ with $0 < p, \mu, \beta < 1$, $w > 0$, $0 < x < \min\{p, (1 - \mu)^w\}$, $\mathcal{Q} \neq \emptyset$, $\text{CERT}(p, \mu, w, \beta, x) < 0$, a source index σ , and $0 < \xi < \min\{x, e^{-A_\sigma}\}$, $0 < y_0 < e^{-B_\sigma}$, $0 < \nu < \mu$. We write $p_\gamma = p$, $w_\gamma = w$, and $A_\gamma = -\log \xi$, $B_\gamma = -\log y_0$, $b_\gamma = -\log \nu$.

A control packages everything Theorem 4.2 needs, with $\hat{x} = e^{-A_\sigma}$ and $\hat{y} = e^{-B_\sigma}$. The source index says which source claim supplies the stopping rule.

Definition 5.3 (profiles). A *profile* $P = (p_P, I_P, L_P)$ has a density $p_P \in (0, 1)$, a compact interval $I_P \subset (0, \infty)$, and a positive continuous piecewise-affine function L_P on I_P with finitely many rational knots. With constant $C \geq 1$, it stands for the claim

$$(\text{ASSERT}_P) \quad d_R(V) \geq p_P \text{ and } |V| \geq Ce^{kL_P(b/k)} \implies V \supseteq (k, b),$$

for all integers $k, b \geq 1$ with $b/k \in I_P$ and every host V .

A profile is an exponential Ramsey bound for hosts of red density at least p_P , along the ratios $b/k \in I_P$. Note that it is stated at integer pairs (k, b) .

5.2 How profiles are justified

A bank has two kinds of profiles. A *direct* profile is obtained by applying the local step, possibly after a sequence of blue steps. A *combined* profile is obtained by combining direct bounds on the same host, as in the two-bound example. The certificate labels these kinds RAW and CLOSED, respectively. Both kinds are proved by the bank validity theorem; the labels describe their derivations, not their proof status. A *realization* is the data and inequalities that justify a profile on an interval of ratios. We now describe the derivations used in the certificate.

Definition 5.4 (direct realizations). Each direct profile P has a finite cover of I_P by closed intervals J . Each J carries a margin $\varepsilon_J > 0$ and one of the following.

1. *Leaf*. A control γ with $p_\gamma \leq p_P$, such that on J

$$L_P(r) \geq \varepsilon_J + \frac{A_\gamma + r(B_\gamma + w_\gamma b_\gamma)}{1 + w_\gamma}.$$

2. *Retained route*. A control γ with $\pi := p_\gamma < p_P$; a partition $\theta = u_0 < u_1 < \dots < u_M = \omega$ with $\theta > 0$ and $J \subseteq [\theta, \omega]$; for each cell $J_j = [u_{j-1}, u_j]$ a slope d_j and a *child* profile $c(j)$ (any direct or combined profile of the bank) with $J_j \subseteq I_{c(j)}$; and the continuous function ψ on $[\theta, \omega]$ with $\psi(\theta) > 0$ that is affine with slope d_j on J_j . These must satisfy

$$d_j > -\log(1 - p_{c(j)}), \quad \psi(s) > L_{c(j)}(s) \quad (s \in J_j), \quad (12)$$

and, writing A, B, b, w for $A_\gamma, B_\gamma, b_\gamma, w_\gamma$, on J

$$L_P(r) \geq \varepsilon_J + \max\left(\psi(r), \frac{A + rB + w(\theta b + \psi(r) - \psi(\theta))}{1 + w}\right).$$

A leaf applies Theorem 4.2 directly to a balanced cut. A route first spends blue steps along the ratio interval $[\theta, r]$, at the cost recorded by ψ , and applies a child profile whenever the current set is dense enough in red. Only when the blue target has dropped to θk does it apply the local step. The density π of the terminal control is strictly below p_P ; the gap pays for keeping only the rows with red density at least π .

Definition 5.5 (combined realizations). Each combined profile Q has a finite cover of I_Q by closed intervals J , each carrying one of the following. Only direct profiles are used in these combinations, and the inequalities are non-strict.

1. *Density copy*. A direct profile R with $p_R \leq p_Q$, $J \subseteq I_R$ and $L_Q \geq L_R$ on J .
2. *Complementary pair*. Direct profiles R, S with $p_R + p_S \leq 1$, $J \subseteq I_R$, $\{1/r : r \in J\} \subseteq I_S$ and $L_Q(r) \geq \max(L_R(r), r L_S(1/r))$ on J .

Definition 5.6 (source covers). For each source pair h fix $0 < \ell_h < H_h$ and $\varepsilon_h, \eta_h > 0$ with

$$\widehat{f}(\lambda) \leq T_h(\lambda) - \varepsilon_h(1 + \lambda) \quad (\lambda \in (0, \ell_h] \cup [H_h, \infty)), \quad (13)$$

and a finite cover of $[\ell_h, H_h]$ by closed intervals J . Each J is either *grounded*, with $\widehat{f} \leq T_h - \eta_h(1 + \lambda)$ on J , or carries two profiles P_R, P_B with $p_{P_R} + p_{P_B} \leq 1$, $J \subseteq I_{P_R}$, $\{1/\lambda : \lambda \in J\} \subseteq I_{P_B}$ and

$$\max(L_{P_R}(\lambda), \lambda L_{P_B}(1/\lambda)) \leq T_h(\lambda) - \eta_h(1 + \lambda) \quad (\lambda \in J).$$

The source covers are where the feedback happens: a source pair may be justified partly by the Lu–Wang function and partly by the bank’s own profiles.

5.3 Validity of a bank

Theorem 5.7 (bank validity). *Consider a bank whose direct profiles, combined profiles and source pairs have realizations and covers as above. Then there is $C_* \geq 1$ such that, with $C = C_*$, every claim (ASSERT_P) and every claim (SOURCE_h) holds for every host.*

The dependencies between profiles and sources may form cycles. The theorem is still true because every dependency is used on a strictly smaller host. The proof is an induction on the number of vertices of the host.

For $C \geq 1$ and a host V , let $\mathcal{P}_V(C)$ be the statement that every bank claim, with constant C , holds on every proper subset $Z \subsetneq V$, under both namings of the colours.

Lemma 5.8 (leaves and routes). *For each leaf or route piece J of a direct profile P there are $K_J \geq 1$ and $D_J \geq 1$, depending only on the data of the piece and not on C , with the following property. Let V satisfy $\mathcal{P}_V(C)$, and fix a naming of the colours. If $k \geq K_J$, $b/k \in J$, $d_R(V) \geq p_P$ and $|V| \geq D_J C e^{k(L_P(b/k) - \varepsilon_J/2)}$, then $V \supseteq (k, b)$.*

Proof. The stopping rule. Let γ be the control of the piece, with source index σ , and put $\hat{x} = e^{-A_\sigma}$, $\hat{y} = e^{-B_\sigma}$. For every $W \subseteq V$ and every $\ell \geq 1$, the host W satisfies $\text{Stop}(C, \hat{x}, \hat{y}, \ell)$: a proper subset $Z \subsetneq W$ is a proper subset of V , and $|Z| \geq C \hat{x}^{-a} \hat{y}^{-\ell} = C e^{A_\sigma a + B_\sigma \ell}$ gives $Z \supseteq (a, \ell)$ by (SOURCE_σ) on Z . The hypotheses of Theorem 4.2 on $(\xi, y_0, \nu, \hat{x}, \hat{y})$ hold by the definition of a control, so it gives a threshold $L_0(\gamma)$.

Leaves ($D_J = 3$). Choose K_J with $\lfloor \alpha k \rfloor \geq \max(1, L_0(\gamma))$ for $k \geq K_J$, where $\alpha = \min J$. Let $r = b/k$ and $L = L_P(r) - \varepsilon_J/2$. Since $|V| \geq 3C e^{kL} \geq 3$, Lemma 2.1 gives a cut $V = X \sqcup Y$ with $d_R(X, Y) \geq d_R(V) \geq p_P \geq p_\gamma$ and $|X|, |Y| \geq C e^{kL}$. Then

$$\log(|X|^w |Y|) \geq (1+w) \log C + (1+w)kL \geq (1+w) \log C + kA_\gamma + bB_\gamma + wbb_\gamma,$$

by the leaf inequality. This is the size hypothesis of Theorem 4.2 with $t = \ell = b \geq L_0(\gamma)$. So $X \cup Y$ has a red K_k , or X or Y has a blue K_b .

Routes. Write $E(r) = \max(\psi(r), (A + rB + w(\theta b + \psi(r) - \psi(\theta)))/(1+w))$ and $\varepsilon = \varepsilon_J/2$, so $L_P(r) - \varepsilon_J/2 \geq E(r) + \varepsilon =: z$. Let $G = \psi + E(r) - \psi(r) + \varepsilon/2$ on $[\theta, r]$, a vertical shift of ψ . Then

$$z - G(r) = \varepsilon/2, \quad z + wG(\theta) - (A + rB + w\theta b) \geq (1+w/2)\varepsilon.$$

Let $V_d = |d_1| + \sum_{j < M} |d_{j+1} - d_j|$, let $\delta_0 > 0$ be the minimum of $\psi - L_{c(j)}$ over the closed cells, let $\eta_0 = \min_j (1 - p_{c(j)} - e^{-d_j}) > 0$, and let $\psi_{\min} = \min \psi > 0$; all are positive by (12). Put $D_J = 3(1 - \pi)/(p_P - \pi)$, and choose K_J so that for $k \geq K_J$: $\lfloor \theta k \rfloor \geq \max(1, L_0(\gamma))$, $k\varepsilon/2 > V_d$, $k\delta_0 > V_d$, and $\eta_0 e^{k\psi_{\min} - V_d} \geq 1$.

Suppose $|V| \geq D_J C e^{kz}$. Take the cut X_0, Y of Lemma 2.1 with $d_R(X_0, Y) \geq p_P$, and let X be the vertices of X_0 with at least $\pi|Y|$ red neighbours in Y . Counting red edges gives $|X| \geq \frac{p_P - \pi}{1 - \pi} |X_0| \geq C e^{kz}$; also $|Y| \geq C e^{kz}$, and every nonempty $Z \subseteq X$ has $d_R(Z, Y) \geq \pi$. Let $\ell = b$ and $n_0 = \lfloor \theta k \rfloor$. For $s \in (\theta, r]$ let $d(s)$ be the slope of ψ on the cell containing s (the left cell at a knot). Define $\Gamma(t) = kG(\theta)$ for $t \leq n_0$ and $\Gamma(t) = kG(\theta) + \sum_{n_0 < \tau \leq t} d(\tau/k)$ for $n_0 < t \leq \ell$. Comparing the sum with the integral, one cell boundary at a time, gives $|\Gamma(t) - kG(t/k)| \leq V_d$ for $n_0 < t \leq \ell$, and $\Gamma(t) \geq k\psi_{\min} - V_d$ throughout.

We claim: for $t = 1, \dots, \ell$ and every $Z \subseteq X$ with $|Z| \geq C e^{\Gamma(t)}$, either $Z \cup Y$ has a red K_k , or Z has a blue K_t , or Y has a blue K_ℓ .

For $t \leq n_0$, the terminal gap gives $\log(|Z|^w |Y|) \geq (1+w) \log C + k(wG(\theta) + z) > (1+w) \log C + kA + \ell B + wtb$, which is the size hypothesis of Theorem 4.2 for (Z, Y) with the control γ and

$\ell \geq n_0 \geq L_0(\gamma)$. For $n_0 < t \leq \ell$, let $s = t/k$ and let c be the child of the cell of s . Then $\Gamma(t) \geq kL_c(s) + k\delta_0 - V_d > kL_c(s)$. If $d_R(Z) \geq p_c$, then (ASSERT $_c$) on Z , which is a proper subset of V , gives $Z \supseteq (k, t)$. Otherwise some vertex $v \in Z$ has more than $(1 - p_c)(|Z| - 1) \geq e^{-d(s)}|Z|$ blue neighbours Z' in Z , using $|Z| \geq 1/\eta_0$. Then $|Z'| \geq Ce^{\Gamma(t-1)}$, and the claim for $t - 1$ applied to Z' , with v added to a blue K_{t-1} , finishes the step.

Finally $\Gamma(\ell) \leq kG(r) + V_d = kz - k\varepsilon/2 + V_d < kz$, so $|X| \geq Ce^{\Gamma(\ell)}$, and the claim with $Z = X$, $t = \ell = b$ gives $V \supseteq (k, b)$. $\square$

Lemma 5.9 (combined profiles on the same host). *Suppose every direct claim holds on the host V , under both namings of the colours. Then every combined claim holds on V .*

Proof. Let $r = b/k$ lie in a piece J of Q , and let $|V| \geq Ce^{kL_Q(r)}$. For a density copy, $d_R(V) \geq p_Q \geq p_R$ and $L_R(r) \leq L_Q(r)$, so (ASSERT $_R$) applies. For a complementary pair, if $d_R(V) \geq p_R$ then (ASSERT $_R$) applies at (k, b) . Otherwise $d_B(V) > 1 - p_R \geq p_S$; renaming the colours, (ASSERT $_S$) at the pair (b, k) has threshold $Ce^{bL_S(k/b)} = Ce^{krL_S(1/r)} \leq Ce^{kL_Q(r)}$. In both orientations the outcome is a red K_k or a blue K_b in the original names. The complementary case does not use $d_R(V) \geq p_Q$. $\square$

Proof of Theorem 5.7. Constants. Let K be an integer with $K \geq K_J$ for every leaf and route piece, and $K\varepsilon_J/2 \geq \log D_J$ for every direct piece. Let M_I be the larger of 1 and the largest right endpoint of all the intervals I_P , and let $c_0 = \min_h \min(A_h, B_h) > 0$. Put

$$C_{\text{prof}} = \max\{R(k, b) : 1 \leq k < K, 1 \leq b \leq M_I(K - 1)\}, \quad C_{\text{small}} = \max_{0 \leq j \leq K-2} j! (2/c_0)^j,$$

and $C_{\text{tail}} = C_\varepsilon$ from Theorem 2.2 with $\varepsilon = \min_h \min(\varepsilon_h, \eta_h)$. Let $C = \max(1, C_{\text{prof}}, C_{\text{small}}, C_{\text{tail}})$.

Grounding. (a) For a profile P and $k < K$ with $b/k \in I_P$, we have $b \leq M_I(K - 1)$, so $|V| \geq Ce^{kL_P(b/k)} \geq C \geq R(k, b)$ gives $V \supseteq (k, b)$ whatever the density. (b) For a source pair h , if $b/k \notin (\ell_h, H_h)$, then (13) gives $\hat{f}(k, b) + \varepsilon(k + b) \leq kT_h(b/k)$, so Theorem 2.2 proves (SOURCE $_h$). The same computation works on grounded cells. (c) If $\min(k, b) < K$, write $j = \min(k, b) - 1$ and $n = \max(k, b)$; then $\binom{n+j-1}{j} \leq (2n)^j \leq j!(2/c_0)^j e^{c_0 n} \leq Ce^{A_h k + B_h b}$, so Erdős–Szekeres proves (SOURCE $_h$).

Induction. Suppose some claim fails, and let V be a host of least order on which some claim fails, under some naming of the colours. Then $\mathcal{P}_V(C)$ holds.

Stage 1: direct claims on V . Let P be direct, $b/k \in I_P$, $d_R(V) \geq p_P$ and $|V| \geq Ce^{kL_P(b/k)}$. If $k < K$, use grounding. Otherwise take the piece $J \ni b/k$. Since $k\varepsilon_J/2 \geq \log D_J$, $|V| \geq D_J Ce^{k(L_P(b/k) - \varepsilon_J/2)}$, and Lemma 5.8 applies. The same holds with the colours renamed.

Stage 2: combined claims on V . By Stage 1 and Lemma 5.9.

Stage 3: source claims on V . By grounding we may assume $k, b \geq K$ and that $\lambda = b/k$ lies in a cell J carrying profiles P_R, P_B . If $d_R(V) \geq p_{P_R}$, then (ASSERT $_{P_R}$) on V applies, since $kL_{P_R}(\lambda) \leq kT_h(\lambda) = A_h k + B_h b$. Otherwise $d_B(V) > 1 - p_{P_R} \geq p_{P_B}$, and (ASSERT $_{P_B}$) with renamed colours at (b, k) applies, since $bL_{P_B}(1/\lambda) = k\lambda L_{P_B}(1/\lambda) \leq A_h k + B_h b$.

So every claim holds on V , a contradiction. Take $C_* = C$. $\square$

Corollary 5.10 (diagonal bound). *Suppose a combined profile Q has $1 \in I_Q$, and the piece of Q containing 1 is a complementary pair. Then $R(k, k) \leq \lceil C_* e^{kL_Q(1)} \rceil$ for every $k \geq 1$.*

Proof. By Theorem 5.7 the direct claims hold on every host, so the proof of Lemma 5.9 applies to every host at (k, k) . For a complementary pair it never uses the density hypothesis of Q . $\square$

6 The certified bank and the proof of Theorem 1.1

6.1 The bank

The certified bank $\mathcal{B}^*$ is a finite file of rational numbers (Appendix A). It records the controls, profiles, and source comparisons needed for Theorem 5.7. We found the data by numerical search, using the bounds produced by one application of the method as inputs to further applications. The proof does not depend on the sequence of searches or on their numerical accuracy. It depends only on the final file: no floating-point number enters the proof, and every inequality below is certified with exact rational or outward-rounded interval arithmetic. Table 1 lists its contents.

object	number
controls	918
$\beta = 999/1000$; $p_\gamma = j/80 - 10^{-6}$ with $p_\gamma \in [0.0125, 0.95]$ $\mu \in [1/20, 4/5]$; $w \in [1/8, 4]$, with $w < 1$ for 648 controls	
source pairs	2713
of which justified by a single tangent of $\hat{f}$	733
of which justified by covers using profiles	1980
Direct profiles (RAW)	1902
leaf pieces / route pieces	9899 / 7284
applications of child profiles in routes	304 695
Combined profiles (CLOSED, including the root)	1003
density-copy pieces / complementary-pair pieces	9908 / 1686

Table 1: The certified bank $\mathcal{B}^*$.

The *root* is a combined profile Q^* with $p_{Q^*} = 1/2$ and $I_{Q^*} = [1, 81/80]$. It has a single piece, a complementary pair using the same direct profile R^* in both orientations, with $p_{R^*} = 1/2$ and $I_{R^*} = [79/80, 81/80]$. Thus $p_{R^*} + p_{R^*} = 1$, and

$$L_{Q^*}(1) = L_{R^*}(1) = 1.252369589401 < z^*.$$

6.2 The checks

The following properties of $\mathcal{B}^*$ were checked by an exact program. It uses rational arithmetic and interval arithmetic with 224-bit outward rounding for log and exp; it rejects floating-point input.

- (C1) *Controls*. For each of the 918 controls: $0 < x < \min\{p, (1 - \mu)^w\}$; the point $q = \mu$ lies in the feasible set $\mathcal{Q}$ (so $\mathcal{Q} \neq \emptyset$); and $\text{CERT}(p, \mu, w, \beta, x) < 0$. The last check splits $(0, 1)$ into 8192 cells, bounds the integrand above on each cell by a certified quantity, and verifies that the resulting upper bound for the integral is negative. (The Lean version uses up to 65 536 cells for 151 of the controls.) Also $\xi < x$, $\nu < \mu$, and $-\log \xi > A_\sigma$, $-\log y_0 > B_\sigma$ for the terminal source.
- (C2) *Direct pieces*. The leaf and route inequalities, with margins $\varepsilon_J > 0$, the density conditions $p_\gamma \leq p_P$ for leaves and $p_\gamma < p_P$ for routes, the route conditions (12), and the inclusions $J_j \subseteq I_{c(j)}$.
- (C3) *Combined pieces*. The density copy and complementary pair inequalities, densities and domain inclusions, including the reflected domains $\{1/r : r \in J\}$.

- (C4) *Sources.* The tail inequalities (13) and the cover inequalities, with positive margins. Comparisons with $\hat{f}$ use tangent lines of $\hat{f}$, which lie above $\hat{f}$ by Lemma 2.3, and enclosures of f and f' computed from the Lu–Wang data.
- (C5) *Root.* Q^* is combined, $1 \in I_{Q^*}$, its piece at 1 is a complementary pair, $L_{Q^*}(1) < z^*$, and $e^{z^*} < 7/2$. The sharper comparison $e^{z^*} \leq 34986238/10000000$ is also kernel-checked in the final Lean corollary.

For the finite comparisons between profiles and tangent lines, a common refinement at their knots and at the reciprocals of knots of reflected profiles makes each compared function affine on every cell. Indeed, if $L(s) = a + bs$ on a cell, then $rL(1/r) = ar + b$ on the reflected cell. Checking these comparisons at cell endpoints therefore proves them on whole intervals. This argument concerns the profile and tangent-line comparisons, not the integral certificate or the original Lu–Wang function f . The smallest margin in the source checks is 10^{-12} ; combined inequalities are allowed to be tight, and the root inequality $L_{Q^*}(1) \geq L_{R^*}(1)$ holds with equality.

6.3 Proof of Theorem 1.1

By (C1)–(C4), $\mathcal{B}^*$ satisfies all hypotheses of Theorem 5.7. In particular, every control satisfies the assumptions of Theorem 4.2. Theorem 5.7 gives a constant $C_* \geq 1$. By (C5) and Corollary 5.10,

$$R(k, k) \leq \lceil C_* e^{kL_{Q^*}(1)} \rceil \leq (C_* + 1)e^{kL_{Q^*}(1)} \quad (k \geq 1).$$

The second inequality uses $L_{Q^*}(1) > 0$. Set $\delta = z^* - L_{Q^*}(1) = 0.000000100001 > 0$ and choose

$$K = \max \left\{ 1, \left\lceil \frac{\log(C_* + 1)}{\delta} \right\rceil \right\}.$$

For $k \geq K$, we have $C_* + 1 \leq e^{\delta k}$, and hence $R(k, k) \leq e^{z^* k} = b_*^k$. The certified comparison $e^{z^*} \leq 34986238/10000000 < 7/2$ gives the remaining eventual bounds. For all $k \geq 1$, taking $C = C_* + 1$ gives $R(k, k) \leq C e^{z^* k}$. $\square$

7 Verification

7.1 Exact numerical checks

The search for the certificate and the proof of its validity are separate. The search uses floating-point arithmetic to find promising parameters. The exact checker then reads the rational data and verifies (C1)–(C5), using outward-rounded enclosures for logarithms and exponentials. Its integral bounds include every cell, including those near the endpoints. It also checks the feasible-set witness $q = \mu$, all domain restrictions, and all positive margins required by Theorem 5.7. A serial run of the checker, without substituting recorded outputs, passed. The Lean development independently proves the finite inequalities needed by the theorem.

7.2 The Lean formalisation

The whole argument has been formalised in Lean 4 [8] with Mathlib [7]. This includes Theorems 3.4, 4.2 and 5.7 and all lemmas used. It also includes the finite checks: the bank is translated into Lean data, and every inequality of (C1)–(C5), including all 918 integral certificates, is checked by the Lean kernel through `decide`. We do not use `native.decide` for these checks: the kernel checks

the proof terms rather than trusting the result of compiled evaluation. The off-diagonal input, Theorem 2.2, and the tangent lines of $\hat{f}$ are taken from Lu and Wang’s Lean development [5]. That development imports `RamseyLean` [6], a Lean 4 formalisation of results from the CGMS argument. These are mathematical dependencies, not merely sources of numerical data.

Related earlier formalisations include Mehta’s Lean 3 proof [10] and Paulson’s Isabelle/HOL proof [11] of the exponential improvement of Campos, Griffiths, Morris and Sahasrabudhe. We cite them as prior formal work; their source code is not imported into this Lean 4 proof.

7.3 Scale and modular checking

The main development uses Lean 4.33.1, while Lu and Wang’s development uses Lean 4.32.1. To obtain a checked chain in a single toolchain, we port the relevant modules to Lean 4.32.1. In that chain every intermediate statement is supplied by its proof. The pinned Mathlib revisions and reproduction instructions are given in Appendix A.

The publication archive contains 82 190 Lean source files with 4 979 699 lines, occupying about 1.82 GB before compression. This count excludes the external libraries. Most of the source consists of generated certificate data and finite checks, rather than handwritten mathematical proofs. The ported theorem closure contains 82 143 modules, supplemented by 42 bridge modules connecting the proof to the off-diagonal input.

We do not ask Lean to check all this source as one enormous expression or one compilation unit. Each integral certificate is divided into cell computations. Separate modules prove the cell bounds, and further modules combine them into bounds for the integral. The finite bank comparisons and the off-diagonal input are also split into modules. Lean checks each module after the modules it imports. The final modules combine these proved inequalities with the general mathematical theorems.

The count 82 143 refers to the ported local proof, not to every imported dependency. The project inputs were also compiled from source in a separate clean build, with the pinned Mathlib and its library dependencies supplied from their cache. We compared the resulting compiled modules with the inputs used by the port. They agreed either byte-for-byte or, for five modules differing in stored source paths, through matching hashes of all declaration types and proofs or values, with identical names and kinds. The latter comparison is hash-based, not byte-for-byte, and the ported local proof was not rebuilt afterwards. Thus verification used separate source builds and comparisons, not one monolithic rebuild.

Lean loads an imported compiled module without checking its proofs again. The source compilations provide the proof checks; manifests and build stamps identify the sources and outputs to which those checks apply. They do not replace proof checking. Appendix A.3 gives the complete build counts, comparisons and statement checks.

7.4 Final statements and axiom checks

In Lean, `ramseyNumber k l` is the least N such that every simple graph on N vertices contains a clique of size k or an independent set of size l . A graph represents the red edges of a colouring, so an independent set represents a blue clique. The sharp theorem `DiagRamsey.diagonal_le_exp_z` states that, for all sufficiently large k ,

$$(\text{ramseyNumber } k \ k : \mathbb{R}) \leq \exp((626184844701/500000000000 : \mathbb{R}) \ k).$$

The corollary `DiagRamsey.diagonal_le_3p4986238_pow` states

$$\exists K : \mathbb{N}, \forall k : \mathbb{N}, K \leq k \implies (\text{ramseyNumber } k \ k : \mathbb{R}) \leq (34986238/10000000 : \mathbb{R})^k.$$

Its proof includes the kernel-checked comparison $e^{z^*} \leq 34986238/10000000$.

The final axiom reports for the sharp theorem and the 3.4986238^k corollary list only

`propext, Classical.choice, Quot.sound.`

These are Lean’s standard axioms for propositional extensionality, classical choice and quotient soundness. The freshly compiled input theorems `DiagRamsey.lu_wang_uniform_source_bound` and `DiagRamsey.lu_wang_lines_P0r12` also use only these three axioms. No additional axiom, including `sorryAx`, occurs in these final theorem reports.

8 Remarks

Tight margins. The certificate is exact, but its margins are small. The source checks have margins near 10^{-12} . The gap between the root profile exponent and z^* is 0.000000100001 , and the difference between 3.4986238 and e^{z^*} is about 6.2×10^{-9} . A reimplementaion must reproduce sufficiently precise enclosures of f ; the source comparisons here use precision better than 10^{-13} .

Effectivity. The constant C_* and the threshold K are not effective, because the constants of Theorem 3.4 come from a compactness argument.

Acknowledgements

We thank Zhipeng Lu and Sichen Wang for making their mathematical argument, rational source data, exact checkers and Lean development publicly available [4, 5]. Their formalised off-diagonal bounds are an essential input to our proof. We also thank the contributors to `RamseyLean` [6], which is imported by that development.

We are grateful to the Lean developers [8] and the Mathlib community [7] for the proof assistant, mathematical library and proof automation on which our formalisation depends. We also acknowledge the contributors to Batteries [9] and Lean’s standard library, which occur among the dependencies. We thank Mehta and Paulson for their earlier formalisations of the exponential Ramsey improvement [10, 11].

A Data and reproduction

The paper, rational certificate, exact checker and Lean sources will be provided at `[repository URL to be inserted]`. The following hashes identify the data and dependencies independently of that location. Paths below are relative to the `diagonal-ramsey` directory.

A.1 Data and pinned dependencies

- Lu–Wang source data `data/source.json`: SHA-256 `f3e4a4eaf3f778ae125466eb6d47647a38fd38430fdcdffe8c5f774f9718c08c`.
- Lu–Wang repository commit: `5ec98f470c43be747ea638f796d1494f9b4c0177`.
- Rational bank certificate: SHA-256 of the uncompressed file `bc3c24cf97cb9684b97f76ab7bf0e57775525d550a5dd1424d563900897442bc`.
- `RamseyLean` revision: `90e87da214701dd6eb3d56a2c7121839d8269d14`.

- Main development: Lean 4.33.1, Mathlib revision `0df444a360eaa60ab8c11dca51a86af692955474`.
- Ported development and Lu–Wang input: Lean 4.32.1, Mathlib revision `520045ab14e26149ee970e2e617ca04b09bde5d6`.
- Publication sources and build manifests: `publication/P0r12/`. The archive includes `SharpBound.lean`, `FinalCheck.lean` and the ported theorem closure.
- SHA-256 of the recorded port build-output hash list: `dba4b8956cf2399570162edfe4759767a294592c50fae92b8b690da6263128e5`.

A.2 Reproducing the modular build

The modular build instructions are in `verify/README.md`; the entry point is `verify/verify.sh`. The default mode, `MODE=single`, regenerates the certificate sources from the frozen bank, checks the source manifest, and compiles the ported local proof under Lean 4.32.1, including all certificate, bank and source-line modules. Here “single” means that each module is built in one toolchain, not that the proof is checked as one large expression. The optional `MODE=full` also builds the Lean 4.33.1 development separately. Both modes can resume matching completed modules. The setup can restore previously compiled input archives. To check these inputs from source, use a clean `.lake` directory with the pinned dependencies, build the Lu–Wang, `RamseyLean`, bridge and shared-definition modules, and compile the generated tangent-line modules separately. The recorded clean build and its comparisons with the inputs used by the port are described below. The publication archive supplies the sharp theorem sources and the instructions for checking them in the ported environment. Both the script and the archive instructions should be read with their pinned toolchain and dependency manifests; the two Lean versions are not interchangeable.

The build scripts schedule ready modules according to their memory requirements and preserve completed build outputs. A stopped build can therefore resume without recomputing every certificate. Generated sources are compared with a recorded manifest, and source/build stamps tie the saved outputs to the sources from which they were checked.

A.3 Recorded builds and comparisons

The publication validation matched all 82 143 closure stamps and the bridge-source manifests. These hashes establish which sources and outputs were used; the mathematical verification comes from the Lean kernel checks.

The separate clean input build compiled 770 modules from source: 741 from the Lu–Wang development, 16 from `RamseyLean`, 11 bridge modules and two shared definitions. It also compiled the 31 tangent-line modules, giving 801 input modules in total. No archived project build outputs were used in this clean build. The pinned Mathlib and its library dependencies were supplied from their cache.

We compared the resulting `.olean` files, which contain Lean’s compiled modules, with those used by the ported proof. Of the 801 files, 796 were byte-identical. The other five differed in their stored absolute source-file paths. For these five modules, a comparison of all 710 declarations gave identical names and kinds, with matching hashes of their types and of their proofs or values. This last comparison is hash-based, not byte-for-byte. The ported local proof was not rebuilt after this comparison. Thus the verification uses separate source builds and comparisons of their outputs, rather than one monolithic rebuild. Lean loads an imported `.olean` without checking its proofs again; source hashes and build stamps alone would not replace the source compilation of these inputs.

The recorded port and publication checks also establish the following:

- all 82143 ported modules compiled, with no unproved `Theorems.*` statement modules remaining in the final dependency closure;
- the text of the 3.5^k statement matches its statement file, and `Composite.lean` elaborates that statement against the composed proof;
- `SharpBound.lean` compiles the sharp exponential theorem and its rounded corollaries, and `FinalCheck.lean` completes the publication check.

The final theorem and input axiom reports are stated in Section 7.

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